

Winston Chan

437-858-8822 | winstonchan333@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Toronto

Toronto, ON, Canada

Bachelor of Science in Computer Science and Physics, Minor in Mathematics GPA: 3.7 August 2022 – May 2026

- Relevant Courses: Multivariable Calculus, Linear Algebra, Theory of Computation, Systems Programming, Data Structures, Intro to Machine Learning, Neural Nets and Deep Learning, Databases, Intro to Artificial Intelligence

TECHNICAL SKILLS

Languages: Python, Java, SQL, C/C++, JavaScript, TypeScript, HTML/CSS, R, Bash

Frameworks: FastAPI, React, Tailwind, Bootstrap

Databases: Relational databases (SQL-based), schema design, query optimization

Developer Tools: Git, GitHub, GitHub Actions, Docker, Linux, Apptainer, VS Code, PyCharm, IntelliJ

Core Competencies: Software Engineering, REST APIs, Backend Systems, Machine Learning, Bayesian Optimization, Data Analysis, CI/CD

Python Libraries: NumPy, pandas, SciPy, Matplotlib, scikit-learn, PyTorch, BoTorch, Ax Platform, librosa, XGBoost

EXPERIENCE

AI Lab Software Developer

May 2025 – Present

Acceleration Consortium

Toronto, ON, Canada

- Designed and deployed backend systems for large-scale Bayesian optimization workflows supporting researchers
- Built REST APIs with FastAPI to orchestrate long-running, asynchronous optimization jobs
- Integrated an SQL database-backed system to persist experiment metadata, parameters, and outcomes
- Designed and executed Bayesian optimization campaigns using Ax Platform, BoTorch, and PyTorch
- Containerized workloads with Apptainer and scheduled jobs with Linux across high-performance computing environments
- Worked with GPU-accelerated workloads, focusing on efficient memory usage and scalable execution

Software Developer

Sep 2023 – Dec 2023

University of Toronto

Toronto, ON, Canada

- Developed a full-stack Java application for real-time stock data ingestion and visualization using REST APIs
- Applied Clean Architecture principles to design a modular and maintainable codebase
- Collaborated in an agile team using Git-based version control workflows
- [Project Repository](#)

Data Analyst

Mar 2025

SDSS Datathon, University of Toronto

Toronto, ON, Canada

- Applied machine learning and statistical models to real-world astronomical datasets to extract insights
- Performed data cleaning, feature analysis, and model evaluation on large scientific datasets
- Collaborated with teammates using Git and GitHub for version control and reproducible analysis
- [Project Repository](#)

PROJECTS

Piano Music Transcription AI | Python, PyTorch

Jun 2025 – Dec 2025

- Developed an end-to-end audio-to-MIDI transcription system for solo piano music
- Designed and trained convolutional and recurrent neural network architectures under severe class imbalance
- Implemented audio preprocessing pipelines using Fourier-based constant-Q transforms
- [Project Repository](#)

Bullet Hell Game | Java, Swing, AWT

Feb 2025

- Designed and implemented a 2D bullet-hell style game in Java
- Built custom parsers to generate levels dynamically from structured input strings
- [Project Repository](#)